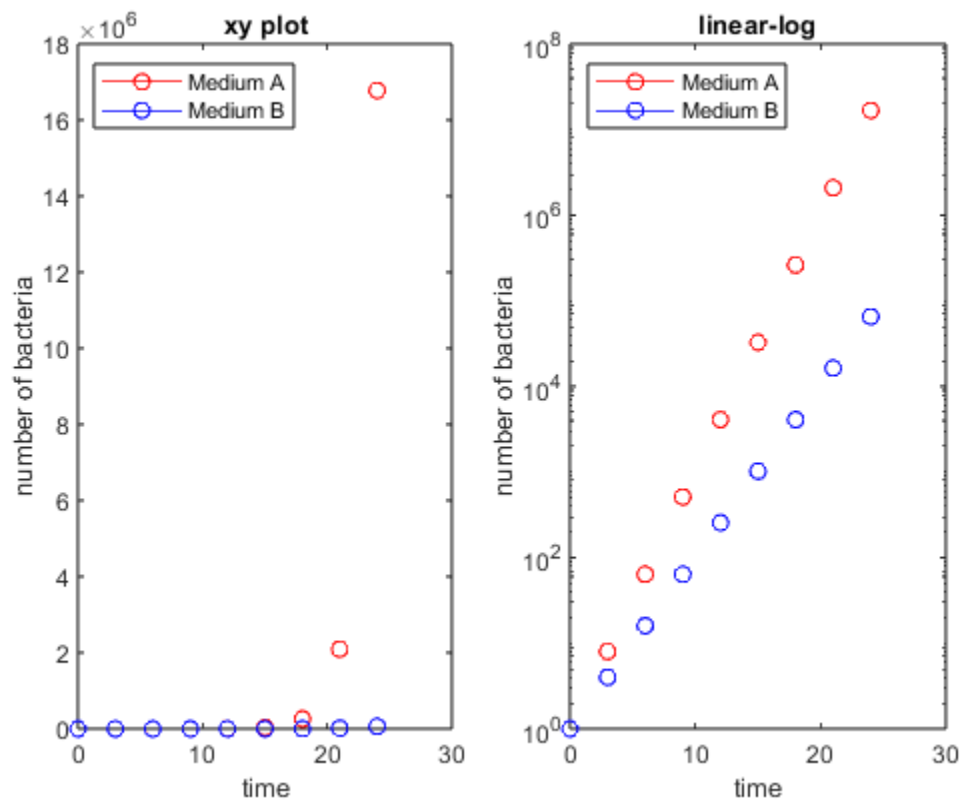

Q1

```
for t=0:3:24
    num_A=2^t;
    num_B=2^(2*t/3);
    subplot(1,2,1)
    plot(t,num_A,'-or')
    hold on
    plot(t,num_B,'-ob')
    xlabel('time')
    ylabel('number of bacteria')
    title('xy plot')
    legend('Medium A','Medium B','Location','northwest')

    subplot(1,2,2)
    semilogy(t,num_A,'-or')
    hold on
    semilogy(t,num_B,'-ob')
    xlabel('time')
    ylabel('number of bacteria')
    title('linear-log')
    legend('Medium A','Medium B','Location','northwest')
end
% The nubmer of bacteria in each media is significantly different
```



Q2

```
MTBFs=[2000 800 3000 5000];  
MTBF=0;  
for i=1:length(MTBFs)  
    MTBF=MTBF+1/MTBFs(i);  
end  
MTBFsys=1/MTBF;  
fprintf('MTBFsys=%f\n',MTBFsys)  
  
MTBFsys=437.956204
```

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